You Could've Discovered the Yoneda Lemma

An exposition on the Yoneda Lemma that's accessible to those without a math background. The Yoneda Lemma is a very useful statement in Category Theory which is usually communicated as "you can completely understand something by observing how everything else relate to it". It has been the subject of intense hype among pseudo-intellectual circles surrounding tech and AI, despite, as we'll see in this post, the statement of the Yoneda Lemma is surprisingly dull.

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1. Introduction

The Yoneda Lemma has gained quite an aura of mystic recently (recently?) as a statement with deep philosophical intent. It's usually phrased in a somewhat abstracted way like "to know thine own self, an individual needeth but seek to comprehend those who surrounds them." or some other vague aphorism. While there is indeed some value to overly abstracted explanations of the Yoneda Lemma, however mystified (or mythified) they are, amongst certain spheres in the vibrant tapestry of human expression lies a group of pseudo-intellectuals who elevate the lemma to, perhaps, a level of divinity? As if the lemma speaks to some sort of spiritual truth.

Take for instance the following examples:

- Relational Quantum Dynamics: A "theory" of um... I don't actually know. Mahayana Buddhism gets mentioned somewhere. Indra's Net seems to be a common theme in these sorts of articles.
- Yon-lang: A.... topos-oriented programming language? What? It mentions the Mathieu group M_{24} and related concepts several times.
- The Relateful Company: A self-help company which has this to say about the Yoneda Lemma.

I want to clarify though, the Yoneda slogan *is legitimate in mathematics* and is beloved by mathematicians. The crackpottery is treating it as a foundational truth of *reality*.

Perhaps this isn't too surprising. Mathematics itself is often mythologised as the Purest of Truth, and those who pursue it as The Intelligents of the Intelligent (I have an upcoming post about this). Category theory, in which the Yoneda Lemma is foundational in, is justly referred to as *the language of mathematics*, and is hence subject to the same mythologising. Afterall, part of the vast utility of

the Yoneda Lemma is because it is an 'obvious' statement in a language with vast applicability in mathematics.

And obvious it is, as I'll posit in this post. With the right pictures in your head, the Yoneda Lemma will seem more akin to a tautology (like *it is what it is*). And this is what I'll be trying to achieve in this article: To put the right pictures in your head so that the Yoneda Lemma seems almost boring. This is, in my opinion, the first step to understanding the applicability and genius of the Yoneda Lemma, as the magic lies not in the statement itself, but in the *language* that makes the statement even possible to state.

This article is written to be understandable even to people who do not have much of a background in mathematics (headingowever, some familiarity with concepts like "sets" and "functions" are required). Hence, I'll be expositing informally, but in great detail, the basic language of Category Theory, but will not elaborate on its vast applicability to mathematics. This is, perhaps, a shame, as the language of category theory cannot be properly appreciated without sufficient mathematical maturity. For the unlikely few who have read this far and are interested in such an exposition, I highly recommend Emily Riehl's Category Theory in Context [1]. Otherwise, this article will not help in a greater appreciation of the Yoneda Lemma for mathematics itself.

2. Basic Definitions

The precise statement of the Yoneda Lemma requires constructions from Category Theory to understand, which I'd be unpacking later in this post. For now, as a bird's eye view, I'll first state the precise statement as typically presented, and summarise at a high level what its saying before diving into the details. So first, the Yoneda Lemma:

Lemma 2.1 (Yoneda)**.** *Let $\mathcal{C}$ be a locally small category. For any functor $F : \mathcal{C} \to \mathrm{Set}$,*

$$\mathrm{Nat}(\mathrm{Hom}(A, -), F) \cong FA$$

naturally in $A \in \mathcal{C}$ and in $F \in [\mathcal{C}, \mathrm{Set}]$

At a high level, the Yoneda Lemma asserts the existence of a certain *bijection* between two sets $\mathrm{Nat}(\mathrm{Hom}(A, -), F)$ and FA, which I'd be describing shortly. This *bijection* is *structure preserving* in some sense, and a consequence of this allows one to capture the "properties" of A as the "set of functions" going out of A, written $\mathrm{Hom}(A, -)$. As Tom Leinster puts it in [2], $\mathrm{Hom}(A, -)$ can be thought of as "how A sees the world".

Regretfully, for illustration purposes, I'd only be expositing on a weaker version of the lemma. In particular, I'll simply be illustrating the *bijection*, disregarding the *structure preservation* (naturality), and only briefly discuss the latter and some of its consequences in Mathematics. This is arguably controversial, as it severly diminishes the utility of the lemma. However, since this article only aims to paint a rough 'picture' of the lemma, I've opted to exclude it, and hopefully this 'picture' can make it 'obvious enough' that the two objects are *the same thing* to believe the lemma's *naturality* clause. A proper exposition of the lemma can be found in [2] and [1]. I've also replaced *locally small* with *small* so I can avoid certain technical issues in the later discussions.

Lemma 2.2 (Yoneda Modified)**.** *Let $\mathcal{C}$ be a small category. For any functor $F : \mathcal{C} \to \mathrm{Set}$, there is a bijection:*

$$\mathrm{Nat}(\mathrm{Hom}(A, -), F) \cong FA$$

With that out of the way, lets proceed with the definitions of the objects we'll be working with.

2.1. Categories

Categories are the central objects of, well, *Category Theory*. Categories can be thought of as mathematical universes which contain objects, of which Category Theory provides an incredibly rich language to specify the relationships between said objects. This rich language is one of the reasons Emily Rielh has called Category Theory a cross-disciplinary language that formalizes mathematical analogies, providing a unified framework to describe the same phenomena in many different areas of Mathematics.

To start this discussion, I'll first provide the formal definition of a category, and breakdown the definition with some examples.

Definition 2.3 (Categories)**.** A *category* consists of:

- A collection of *objects* (which we often denote with uppercase letters $X, Y, Z, ...$)
- A collection of *morphisms* (which we often denote with lowercase letters $f, g, h, ...$)

so that:

- Each morphism has a specified *source* and *destination*. $f : X \to Y$ denotes a morphism with source X and destination Y.
- Each object X has an *identity morphism* denoted $\mathrm{id}_X : X \to X$.
- For any pair of *composable* morphisms $f : X \to Y$ and $g : Y \to Z$, defined so when f's destination is g's source, there exists a composite morphism $f; g : X \to Z$. This composition is also denoted

$$X \underset{f;g}{\longrightarrow} Z = X \underset{f}{\longrightarrow} Y \underset{g}{\longrightarrow} Z$$

This data is subjected to the following constraints:

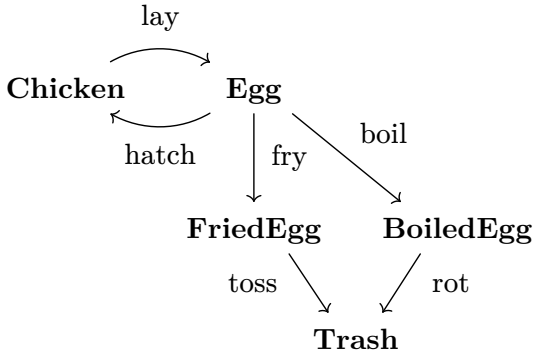
- For any $f : X \to Y$, the composites $\mathrm{id}_X; f$ and $f; \mathrm{id}_Y$ are equal to f. Hence id_X can be thought of as the 'do-nothing' morphism of X.
- For any composable triple of morphisms f, g, h, the composites $(f; g); h$ and $f; (g; h)$ are equal, and, since the ambiguity is addressed, will simply be denoted $f; g; h$.

To concretise above, I'll go through an example toy category I'll call **ChickenAndEgg**.

Example. **ChickenAndEgg** is a category that contains objects

$$\mathrm{Obj} = \{\mathbf{Chicken}, \mathbf{Egg}, \mathbf{FriedEgg}, \mathbf{BoiledEgg}, \mathbf{Trash}\}$$

and the following morphisms and its compositions:



Where we (arbituary) assert that $\mathrm{boil}; \mathrm{rot} = \mathrm{fry}; \mathrm{toss}$. This is representative of how we've arbituary decided not to distinguish **Trash** originating from a **FriedEgg** or **BoiledEgg**.

Note that the diagram in the above Example does not show all morphisms in **ChickenAndEgg**. In particular, I do not draw any of the 'do-nothing' morphisms (id_X for $X \in \mathrm{Obj}$). Further, I do not

include any of the 'compositions'. For instance, $\mathtt{lay}; \mathtt{boil}$ is a morphism in $\mathbf{ChickenAndEgg}$ via Definition 2.3 (Categories) but is not drawn.

Also, note that even though two morphisms can have the same source and destination, they might not be equal. For instance, $\mathtt{lay}; \mathtt{hatch}$ and $\mathrm{id}_{\mathbf{Chicken}}$ both have their source and destination as $\mathbf{Chicken}$ but are not considered equal in this category, perhaps since we would want to distinguish between chickens originating from different eggs.

2.2. Functors

2.3. Natural Transforms

2.4. Hom-Functors

3. The Yoneda Lemma

3.1. The Yoneda Slogan

Bibliography

[1] E. Riehl, *Category theory in context*. Mineola, New York: Dover Publications, 2017.

[2] T. Leinster, *Basic Category Theory*, vol. 143. in Cambridge Studies in Advanced Mathematics, vol. 143. Cambridge University Press, 2014. doi: 10.1017/CBO9781107360068.